MTH202 - Discrete Mathematics (Quiz 3)

Question # 7 of 10 (Start time: 02:11:52 PM, 04 July 2025)

The Product of an even and odd integer is .....

Select the correct option

●	even
○	odd

MTH202 - Discrete Mathematics (Quiz 3)

Question # 4 of 10 (Start time: 02:17:25 PM, 04 July 2025)

If 'n' is an odd integer then n^3+n is .....

Select the correct option

●	even
○	odd

MTH202 - Discrete Mathematics (Quiz 3)

Question # 5 of 10 (Start time: 02:11:29 PM, 04 July 2025)

If 'n' is an odd integer then '3n+2' is ....

Select the correct option

- even
- odd

MTH202 - Discrete Mathematics (Quiz 3)

Question # 9 of 10 (Start time: 02:23:20 PM, 04 July 2025)

If 'n' is an odd integer then '3n+2' is ....

Select the correct option

○	even
◉	odd

MTH202 - Discrete Mathematics (Quiz 3)

Question # 2 of 10 (Start time: 02:21:34 PM, 04 July 2025)

The sum of two odd integers is ....

Select the correct option

- (●) even
- ( ) odd

MTH202 - Discrete Mathematics (Quiz 3)

Question # 4 of 10 (Start time: 02:11:17 PM, 04 July 2025)

A predicate is a sentence that contains a finite number of ----------

Select the correct option

- (●) variables
- ( ) Constants

MTH202 - Discrete Mathematics (Quiz 3)

Question # 9 of 10 (Start time: 02:12:28 PM, 04 July 2025)

Proof by Contradiction is a/an..... method of proof.

Select the correct option

○	Direct
◉	Indirect

MTH202 - Discrete Mathematics (Quiz 3)

Question # 3 of 10 (Start time: 02:17:06 PM, 04 July 2025)

(-1)^n = 1 for n=.....
(^ stands for power)

Select the correct option

○	7
○	5
○	3
●	4

MTH202 - Discrete Mathematics (Quiz 3)

Question # 2 of 10 (Start time: 02:10:47 PM, 04 July 2025)

The sum of any rational number and any irrational number is irrational.

Select the correct option

○	False
◉	True

MTH202 - Discrete Mathematics (Quiz 3)

Question # 3 of 10 (Start time: 02:11:05 PM, 04 July 2025)

The division by zero is allowed in mathematics.

Select the correct option

○	True
◉	False

MTH202 - Discrete Mathematics (Quiz 3)

Question # 7 of 10 (Start time: 02:22:45 PM, 04 July 2025)

The sum of $(6 - 7\sqrt{2})$ and $(6 + 7\sqrt{2})$ is .....

Select the correct option

●	12
○	$12-14\sqrt{2})$
○	$2(6+7\sqrt{2})$
○	$2(6-7\sqrt{2})$

MTH202 - Discrete Mathematics (Quiz 3)

Question # 10 of 10 (Start time: 02:12:42 PM, 04 July 2025)

The product of any two consecutive positive integers is divisible by 2.

Select the correct option

- (●) True
- ( ) False

MTH202 - Discrete Mathematics (Quiz 3)

Question # 1 of 10 (Start time: 02:21:09 PM, 04 July 2025)

The statement $n^2 > n + 3$ is true when _____ .

Select the correct option

○	$n \geqslant 2$
○	$n \geqslant -1$
●	$n \geqslant 3$
○	$n \geqslant 1$

MTH202 - Discrete Mathematics (Quiz 3)

Question # 6 of 10 (Start time: 02:22:31 PM, 04 July 2025)

The sum of any rational number and any irrational number is irrational.

Select the correct option

◉	True
○	False

MTH202 - Discrete Mathematics (Quiz 3)

Question # 1 of 10 (Start time: 02:16:15 PM, 04 July 2025)

In inductive property the first iteration of the loop is solved for?

Select the correct option

○	None
○	I(k)
●	I(0)
○	I(k + 1)

MTH202 - Discrete Mathematics (Quiz 3)

Question # 6 of 10 (Start time: 02:11:39 PM, 04 July 2025)

The ---- of a predicate variable is the set of all values that may be substituted in place of the variable.

Select the correct option

- (●) Domain
- ( ) Range

MTH202 - Discrete Mathematics (Quiz 3)

Question # 8 of 10 (Start time: 02:18:27 PM, 04 July 2025)

Which of the following is true for n = 1?

Select the correct option

○	2^(2n) - 1 by 2
●	2^(2n) - 1 by 3
○	2^(2n) - 1 by 5
○	2^(2n) - 1 by 4

MTH202 - Discrete Mathematics (Quiz 3)

Question # 10 of 10 (Start time: 02:23:32 PM, 04 July 2025)

The predicate describing the initial state is called _____.

Select the correct option

○	None
○	post-condition of the algorithm
●	pre-condition of the algorithm
○	Out put variables

MTH202 - Discrete Mathematics (Quiz 3)

Question # 9 of 10 (Start time: 02:18:41 PM, 04 July 2025)

Proof by .....uses the equivalence p →q≡ -q→-p.

Select the correct option

◉	contraposition
○	contradiction
○	mathematical induction
○	superposition

MTH202 - Discrete Mathematics (Quiz 3)

Question # 2 of 10 (Start time: 02:16:27 PM, 04 July 2025)

The statement $4^n > 3^n + 4$ is true when _____ .

Select the correct option

○	$n = 0$
●	$n \geqslant 2$
○	None of the above
○	$n = 1$

MTH202 - Discrete Mathematics (Quiz 3)

Question # 10 of 10 (Start time: 02:18:56 PM, 04 July 2025)

Which of the following is not an irrational number?

Select the correct option

●	$\sqrt{4}$
○	$\sqrt{5}$
○	$\sqrt{3}$
○	$\sqrt{2}$

MTH202 - Discrete Mathematics (Quiz 3)

Question # 5 of 10 (Start time: 02:17:46 PM, 04 July 2025)

In division algorithm the pre-condition is?

Select the correct option

- (●) a is a nonnegative integer and d is a positive integer
- ( ) q and r are nonnegative integers

MTH202 - Discrete Mathematics (Quiz 3)

Question # 5 of 10 (Start time: 02:22:15 PM, 04 July 2025)

Proof by .....uses the equivalence $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

Select the correct option

○	superposition
○	mathematical induction
○	contradiction
◉	contraposition

MTH202 - Discrete Mathematics (Quiz 3)

Question # 3 of 10 (Start time: 02:21:45 PM, 04 July 2025)

Which of the following is not an irrational number?

Select the correct option

○	√5
○	√2
○	√3
◉	√4

MTH202 - Discrete Mathematics (Quiz 3)

Question # 1 of 10 (Start time: 02:10:34 PM, 04 July 2025)

A predicate becomes ---------- when its variables are given specific values.

Select the correct option

○	algorithm
○	iteration
○	statement
◉	sentence

MTH202 - Discrete Mathematics (Quiz 3)

Question # 7 of 10 (Start time: 02:18:13 PM, 04 July 2025)

Proof by .....uses the equivalence $p \rightarrow q \equiv (p \wedge \neg q) \rightarrow c$

Select the correct option

○	mathematical induction
○	superposition
○	contraposition
●	contradiction

MTH202 - Discrete Mathematics (Quiz 3)

Question # 8 of 10 (Start time: 02:12:15 PM, 04 July 2025)

Proof by contraposition is based on the logical equivalence between a statement and its..... .

Select the correct option

○	none of these
◉	contrapositive
○	Inverse
○	Converse

MTH202 - Discrete Mathematics (Quiz 3)

Question # 8 of 10 (Start time: 02:22:59 PM, 04 July 2025)

The contrapositive of "5n + 2 is odd, then n is odd" is .....

Select the correct option

○	" if n is even then 5n + 2 is not even"
○	" if n is not even then 5n + 2 is even"
●	" if n is even then 5n + 2 is even"
○	" if n is not even then 5n + 2 is not even"

MTH202 - Discrete Mathematics (Quiz 3)

Question # 4 of 10 (Start time: 02:22:01 PM, 04 July 2025)

To prove by mathematical induction for 1+5+9+----------+(4n-3)=n(2n-1) for all positive integers,the basis step is _____.

Select the correct option

○	1=1(2(1)+1)
◉	1=1(2(1)-1)

Cl

MTH202 - Discrete Mathematics (Quiz 3)

Question # 6 of 10 (Start time: 02:18:01 PM, 04 July 2025)

Which of the following is not a method to prove the mathematical statements?

Select the correct option

○	Proof by contraposition
○	Mathematical induction
●	Permutation
○	Proof by contradiction